Theory of Computation $\qquad \qquad \qquad 1-7 \qquad \qquad \qquad$ Review of Mathematical Theory

Let us discuss each one by one -

i) Negation

This proposition is also called NOT proposition. It is denoted by $\neg$ or $\sim$ or having horizontal bar on letter.

If S is a proposition then $\neg S$ defines NOT S. The truth table for this connective is

S	$\neg S$
T	F
F	T

ii) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by $A \land B$. The truth table for this connective is

A	B	$A \land B$
T	T	T
T	F	F
F	T	F
F	F	F

iii) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by $A \lor B$. The truth table for this connective is

A	B	$A \lor B$
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted by $A \to B$ or $A \implies B$.

For Semester - VI (CE / CSE) ...



Figure: Collage of book covers for Software Engineering (A. A. Puntambekar), Theory of Computation (A. A. Puntambekar), Advanced Java (A. A. Puntambekar), Web Technology (A. A. Puntambekar), Computer Graphics (A. P. Godse, Dr. D. A. Godse), Embedded & VLSI Design (V. S. Bagad, A. P. Godse), and Distributed Operating System (I. A. Dhotre)



Figure: Circular graphic with numbers 1 to 5 detailing app features: FREE MODEL PAPER AS PER NEW PATTERN, NO NEED OF INTERNET WHILE USING APP, SYLLABUS & UNIVERSITY PAPERS, IMPORTANT POINTS TO REMEMBER, FORMULAE AT A GLANCE, surrounding the central text GET FREE ANDROID APP DECODE SEM VI CSE

Distributors

* **UBS Publishers Pvt Ltd**

Shop No.133-134, Daryapur Gate, Ahmedabad,

Gujarat-6, Ph: (079) 22160371 / 372 / 373,

email: ubspdahm@ahm.ubspd.com

* **Jaico Publishing House**

2, Mill Officers Colony, Ashram Road, Ahmedabad.

email: jaicoahm@yahoo.co.in

* **Mahajan Book Depot**

Gandhi Road, Ahmedabad - 1.

Ph: (079) 25356031.

* **Patel Book Agency**

1st Floor, Mahavir Smruti Complex, Gandhi Road,

Ahmedabad-1, Ph: (079) 25324741.

* **Atul Agencies**

Fernandis Bridge, Gandhi Road,

Ahmedabad - 1, Ph: (079) 22160475.

Or Contact

* **Ajay Pachegaonkar (Technical Publications)**

Mkt. Executive (Gujarat)

Mob No.: 09687118452,

email: ajay@vtubooks.com

ONLINE SELLERS



Figure: QR Code and logo for Technical Publications

amazon.in

flipkart.com

Price : ₹ 180/-

ISBN 978-93-332-0793-5

$$\text{9}\Vert\text{789333}\Vert\text{207935}$$



Figure: Technical Publications Logo

TECHNICAL PUBLICATIONS

An Up-Thrust for Knowledge

* ![Figure: Facebook icon](figure_placeholder) https://www.facebook.com/technicalpublications
* ![Figure: Twitter icon](figure_placeholder) https://twitter.com/p_technical
* ![Figure: Google+ icon](figure_placeholder) https://plus.google.com/102423981228640599931/posts
* ![Figure: LinkedIn icon](figure_placeholder) http://www.linkedin.com/company/technical-publications
* ![Figure: Pinterest icon](figure_placeholder) http://www.pinterest.com/technicalebooks

Theory of Computation

Regular Languages

Consider the output sequence for Moore machine for input sequence,

$$a\ b\ b\ a$$

Then

$$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{b} q_1 \xrightarrow{a} q_0$$

$$0 \quad\quad 0 \quad\quad 1 \quad\quad 1 \quad\quad 0$$

Similarly output sequence for Mealy machine will be

$$q_0 \xrightarrow{a/0} q_1 \xrightarrow{b/1} q_1 \xrightarrow{b/1} q_1 \xrightarrow{b/0} q_0$$

We can note that length of Moore machine is $n+1$ and that of Mealy machine is n.

13.3 Method for Conversion of Mealy Machine to Moore Machine

In Moore machine the output is associated with every state and in Mealy machine the output is given along the edge with input symbol. To convert Moore machine to Mealy machine state output symbols are distributed to input symbol paths. But while converting Mealy machine to Moore machine we will create a separate state for every new output symbol and according incoming and outgoing edges are distributed.

> Let $M = (Q, \Sigma, \Delta, \delta, \lambda, q_0)$ be a Mealy machine then there exists a Moore machine M' equivalent to M.
> The M' can be given as $M' = (Q \times \Delta, \Sigma, \Delta, \delta', \lambda', [q_0, b_0])$ where b_0 is an output symbol selected from Δ. We will calculate δ' and λ' as follows -

$$\delta'([q, b], a) = [\delta(q, a), \lambda(q, a)]$$

$$\lambda'([q, b]) = b$$

δ' defines the move made by M' on input a.

M' on input a.

λ' defines the output made for the corresponding state q.

Thus we have to calculate δ' and λ' for $q_0, q_1, q_2, \dots \dots q_n$ on input $a_0, a_1, \dots \dots a_n$ and should emit the outputs $b_1, b_2, b_3, \dots \dots \dots b_n$.

Let us understand this method of conversion with the help of some examples.

Example 2.80: *Convert the following Mealy machine into equivalent Moore machine.*



Figure: A Mealy machine transition diagram with two states q_0 and q_1. State q_0 has a self-loop labeled $0/A$, and a transition to q_1 labeled $1/B$. State q_1 has a self-loop labeled $0/B$, and a transition back to q_0 labeled $1/A$.

Fig. 2.69